

Mathematics

Natural Exponential Function

Exponential of base e (Napierian)

The explicit form of f

The exponential function of base e is the function f defined over $] - \infty; + \infty[$ by $f(x) = e^x$

such that $e^0 = 1$, $e^1 = e$ and $e^{-1} = \frac{1}{e}$

For every real number x, $e^x > 0$

Property 1:

Let $a \in R$ and $b \in R$

$$* e^a \times e^b = e^{a+b}$$

$$* \frac{e^a}{e^b} = e^{a-b}$$

$$* e^{-a} = \frac{1}{e^a}$$

$$* (e^a)^b = e^{ab}$$

Examples (1):

Calculate and simplify:

$$* e^2 \times e^3 = e^5$$

$$* (e^x)^2 = e^{2x}$$

$$* \frac{e^{3x+4}}{e^{x+2}} = e^{2x+2}$$

Property 2:

- * $e^{\ln x} = x$ for $x > 0$
- * $\ln e^x = x$ for $x \in \mathbb{R}$

Examples (2):

Calculate and simplify:

$$* e^{2+\ln 3} = e^2 \cdot e^{\ln 3} = 3e^2$$

$$* e^{\frac{1}{2}-\ln 3} = e^{\frac{1}{2}} \cdot e^{-\ln 3} = e^{\frac{1}{2}} \cdot e^{\ln \frac{1}{3}} = \frac{1}{3} \sqrt{e}$$

Equations and in- equalities:

Let $a \in R$ and $b \in R$

- * $e^a = e^b$ then $a = b$
- * $e^a > e^b$ then $a > b$
- * $e^a < e^b$ then $a < b$

Examples:

Solve the following:

a) $e^{3x+2} = e^5$

$$3x + 2 = 5$$

$$x = 1$$

b) $e^x + 3e^{-x} = 4$

$$e^x(e^x + 3e^{-x}) = 4e^x$$

$$e^{2x} - 4e^x + 3 = 0$$

Let $e^x = A$ where $A > 0$

$$e^{2x} - 4e^x + 3 = 0$$

$$A^2 - 4A + 3 = 0$$

$$A = 1 \text{ and } A = 3$$

$$e^x = 1 \text{ then } x = 0$$

$$e^x = 3 \text{ then } x = \ln 3$$

c) $e^{x-1} \geq e^{-2}$

$$x \geq -1$$

d) $e^x \leq 2$

$$x \leq \ln 2$$

Derivative of natural exponential function

- * The exponential function of base e is $f(x) = e^x$ defined over $] - \infty; + \infty[$

The derivative of the Naperian exponential function is very special $f'(x) = e^x$

- * The exponential function of base e is $f(x) = e^U$ defined over $] - \infty; + \infty[$

The derivative of the Naperian exponential function is $f'(x) = U' e^U$

Examples

Find the derivative function of the following :

$$* f(x) = e^{x^2+3x}$$

then

$$f'(x) = (2x + 3)e^{x^2+3x}$$

$$* f(x) = \frac{e^x}{x^2+1}$$

then

$$f'(x) = \frac{e^x(x^2 + 1) - (2x)e^x}{(x^2 + 1)^2} = \frac{e^x(x - 1)^2}{(x^2 + 1)^2}$$

Solved exercises about derivative

Calculate the derivative function of each of the following

1) $f(x) = e^{x^2+1}$

$$f'(x) = (x^2 + 1)' e^{x^2+1} = 2x e^{x^2+1}$$

2) $f(x) = e^{-x+1}$

$$f'(x) = -e^{-x+1}$$

3) $f(x) = x e^x = uv$

$$f'(x) = x' e^x + x(e^x)' = e^x + x e^x = (1 + x) e^x$$

$$4) f(x) = (-x + 4)e^{-x}$$

$$u = -x + 4 \quad u' = -1$$

$$v = e^{-x} \quad v' = -e^{-x}$$

$$\begin{aligned} f'(x) &= u'v + v'u = -e^{-x} + (-x + 4)(-e^{-x}) \\ &= -e^{-x} - (-x + 4)e^{-x} \\ &= e^{-x}(-1 - (-x + 4)) \\ &= e^{-x}(-1 + x - 4) \\ &= (x - 5)e^{-x} \end{aligned}$$

$$5) f(x) = \frac{1+e^{-x}}{1+e^x}$$

$$u = 1 + e^{-x}$$

$$u' = -e^{-x}$$

$$v = 1 + e^x$$

$$v' = e^x$$

$$f'(x) = \frac{-e^{-x}(1 + e^x) - e^x(1 + e^{-x})}{(1 + e^x)^2}$$

$$f'(x) = \frac{-e^{-x} - 1 - e^x - 1}{(1 + e^x)^2} = \frac{-2 - e^{-x} - e^x}{(1 + e^x)^2} = \frac{-2 - \frac{1}{e^x} - e^x}{(1 + e^x)^2}$$

Limits at $\pm\infty$

- * $\lim_{x \rightarrow -\infty} e^x = 0$
- * $\lim_{x \rightarrow +\infty} e^x = +\infty$
- * $\lim_{x \rightarrow +\infty} \frac{e^x}{x} = +\infty$
- * $\lim_{x \rightarrow +\infty} \frac{e^x}{x^\alpha} = +\infty$ (where α is a positive rational number)
- * $\lim_{x \rightarrow -\infty} x e^x = 0$
- * $\lim_{x \rightarrow -\infty} x^\alpha e^x = 0$ (where α is a positive rational number)

Exercise about limits

$$1) \lim_{x \rightarrow -\infty} x^2(e^x + 1) = (-\infty)^2(e^{-\infty} + 1) = +\infty \times (0 + 1) \\ = +\infty$$

$$2) \lim_{x \rightarrow -\infty} -2xe^x = -2(-\infty)e^{-\infty} \\ = +\infty \times 0 \text{ undetermined form}$$

Since e^x is faster than x then

$$\lim_{x \rightarrow -\infty} -2xe^x = \lim_{x \rightarrow -\infty} -2e^x = -2e^{-\infty} = -2 \times 0 = 0$$

$$3) \lim_{x \rightarrow +\infty} (x+1)e^{-x} = +\infty \times e^{-\infty} = +\infty \times 0$$

$$\lim_{x \rightarrow +\infty} (x+1)e^{-x} = \lim_{x \rightarrow +\infty} \frac{x+1}{e^x} = \frac{+\infty}{e^{+\infty}} = \frac{+\infty}{+\infty}$$

By applying l'hôpital rule

$$\lim_{x \rightarrow +\infty} (x+1)e^{-x} = \lim_{x \rightarrow +\infty} \frac{x+1}{e^x} = \lim_{x \rightarrow +\infty} \frac{(x+1)'}{(e^x)'} = \lim_{x \rightarrow +\infty} \frac{1}{e^x} = \frac{1}{e^{+\infty}} = \frac{1}{+\infty} = 0$$

Y=0 H.A

Variation of the function f defined by $f(x)=e^x$

Let f be the function defined over $]-\infty; +\infty[$ by $f(x) = e^x$

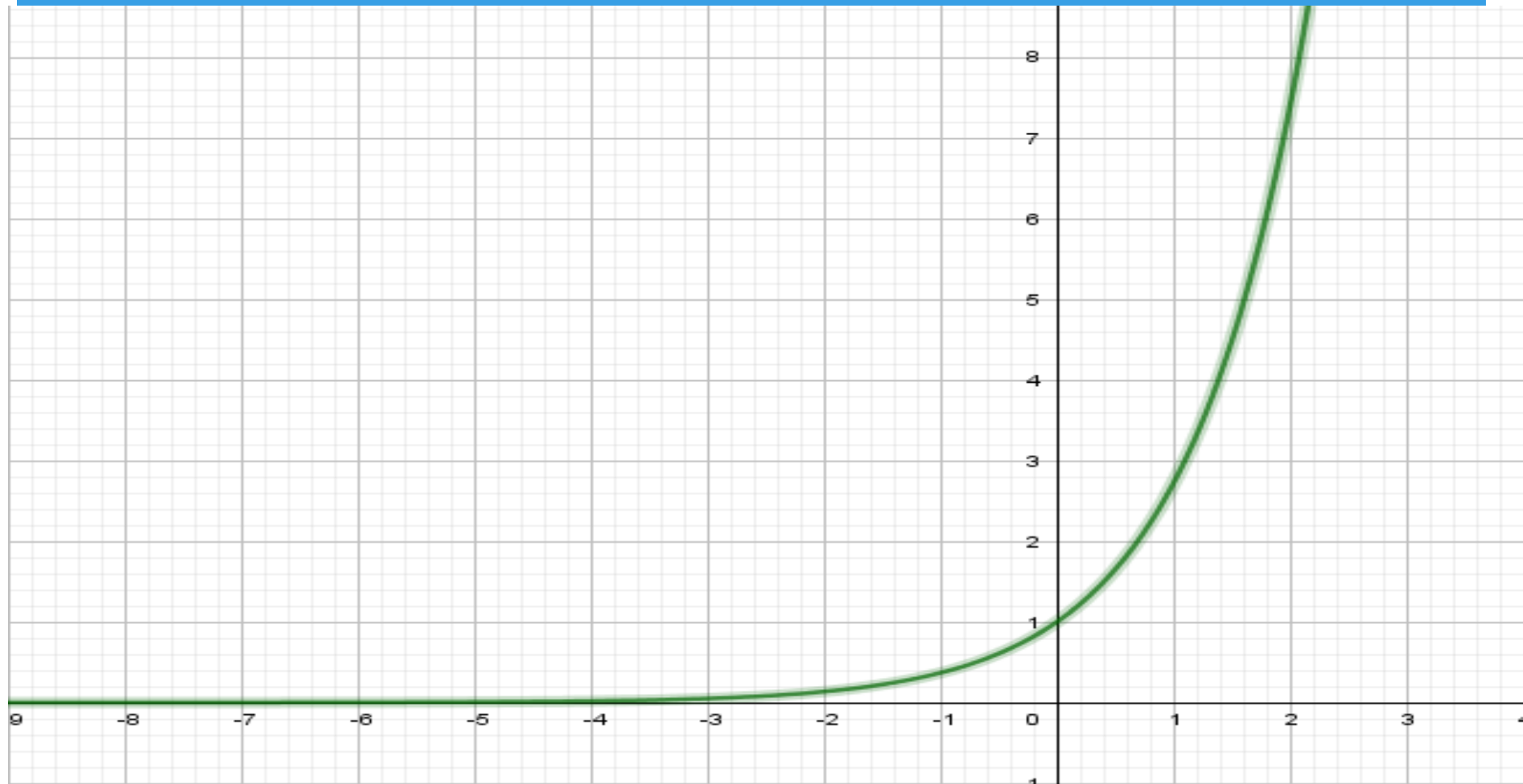
$$\lim_{x \rightarrow -\infty} e^x = e^{-\infty} = 0 \text{ so } y = 0 \text{ H.A.}$$

$$\lim_{x \rightarrow +\infty} e^x = e^{+\infty} = +\infty$$

$f'(x) = e^x > 0$ so f is an increasing function

x	$-\infty$	$+\infty$
$f'(x)$	+	
$f(x)$	0	$+\infty$

Representative curve of e^x



Natural Exponential Function Links

<https://youtu.be/XNUvePqFCzo>

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Thank You